

Assignment 2 – NLP (Chat LLM)

Assignment Objective

Develop a Natural Language Processing (NLP) prototype based on **Prompt Engineering** for interaction with a **chat-oriented Large Language Model (LLM)**. The assignment should cover the full development process, from the conceptual definition of the product to the formal design of the prompt, its practical implementation, testing, and critical evaluation.

1. Product Purpose

- Define the problem the prototype is intended to solve.
- Explain the relevance of the proposed prompt-based solution.
- Identify the potential users, domains, or practical applications of the prototype.

2. Model or Method (Prompt)

- Explain the theoretical foundation of the Prompt Engineering approach adopted in the project.
- Justify the choice of the LLM and the chat environment used for implementation.
- Present the main techniques applied in prompt development, such as role assignment, task specification, contextual grounding, step structuring, output constraints, or other relevant strategies.
- Explain how the prompt design supports the intended objective of the prototype.

3. Solution Design (Formal Prompt Description)

- Describe the formal structure of the prompt, including its main components and variables.
- Explain how the instructions were constructed and refined.
- Present the strategies used to improve response quality, consistency, relevance, and task alignment.
- Clarify how the prompt design guides the behavior of the LLM within the proposed application.

4. Implementation (Operational Form of the Prompt)

- Present the operational version of the prompt as used in the chat environment.
- Provide examples of prompt use in realistic interaction scenarios.
- Show adjustments, refinements, or iterations made during development.
- Demonstrate variations of the prompt and explain their impact on the generated responses.

5. Product Functionality Testing

- Record and present different LLM interactions using the developed prompt.
- Analyze the quality, coherence, and adequacy of the generated responses.

- Discuss the strengths and limitations of the current prompt design.
- Suggest possible improvements, refinements, or future extensions.

6. Development Experience

- Report the main challenges encountered during prompt design and testing.
- Explain how these challenges were addressed or mitigated.
- Reflect on the knowledge, competencies, and practical insights acquired throughout the project.

7. Market Potential

- Indicate whether the proposed product could be useful in a specific area, sector, or market.
- Present realistic scenarios in which the solution could be applied.
- Discuss possible beneficiaries, use cases, or opportunities for future development.

Submission

The assignment must be submitted in **PDF report format** and should include all stages defined in this guideline, from conceptual framing to implementation, testing, and final evaluation.

Evaluation Criteria

The assignment will be evaluated according to the following criteria:

- Creativity and innovation in the developed solution.
- Clarity in the presentation of the problem and proposed solution.
- Coherence in the description of the prompt-based model or method.
- Quality of prompt design and documentation.
- Quality of testing, analysis, and evaluation of results.